Fee Concentration Insurance

Mathematical Specification

ThetaSwap

March 5, 2026

Abstract

We specify an insurance mechanism against adverse fee concentration in Uniswap V4 liquidity provision. The fee concentration index $A_T = (\sum_k \theta_k \cdot x_k^2)^{1/2}$ measures realized fee inequality across LP positions. The Ma–Crapis competitive null $A_T^{1/N} = \sqrt{\Theta/N^2}$ provides the equal-share baseline. The concentration deviation $\Delta^+ = \max(0, A_T - A_T^{1/N})$ captures excess concentration above the competitive equilibrium and maps to price space via the odds ratio $p = \Delta^+/(1 - \Delta^+)$.

Econometric identification (quadratic deviation exit hazard model) establishes an inverted-U relationship: below the turning point $\Delta^* \approx 0.09$, concentration is protective (shelter effect); above it, concentration drives LP exits (Capponi regime). This turning point calibrates insurance demand.

The hook tracks three co-primary state variables—A_T, Θ, N—with Δ^+ derived on the fly. The mechanism is perpetual with jump-adjusted funding and USDC numeraire.

Cross-pool empirical analysis (Spearman $\rho(A_T, \log \mathrm{TVL}) = +0.19$, weak) confirms no TVL–concentration correlation, selecting the **hybrid** architecture: hook-based mutual insurance with per-pool reserves funded by γ% fee redirect, payouts via `donate()` when $\Delta^+ > \Delta^*$, and ERC-6909 transferable claims bridging to an optional CFMM market layer.

Contents

1 Fee Concentration Index **3**

1.1 Primary Index: A_T 3

1.2 Co-Primary State Variables 3

1.3 Ma–Crapis Competitive Null: $A_T^{1/N}$ 4

1.4 Concentration Deviation: Δ^+ 5

1.5 Price Mapping 5

2 Econometric Calibration **5**
2.1 Quadratic Deviation Model 6
2.2 Key Results 6
2.3 Inverted-U and the Insurance Trigger 6

3 Settled Design Properties **6**
3.1 Backtest-Confirmed Properties 6
3.2 Optimal Payoff Formula: Power-Squared Exit Payoff 7
3.3 Why Power-Squared Dominates 8

4 Architectural Design **8**
4.1 Participants and Token Mapping 8
4.2 LP Payoff (Underwriter) 9
4.3 Borrower Payoff (PLP) 9
4.4 Market-Implied Strike 9
4.5 Connection to PowerMaker 9

5 Trading Function **10**
5.1 Reserve Derivation 10
5.2 Coordinate Transform: $u = x^2$ 10
5.3 Spot Price 11
5.4 Swap Mechanics 11
5.5 Liquidity Provisioning 12

6 Reserve Analysis **12**
6.1 Reserve Curves in Price Space 12
6.2 Monotonicity 12
6.3 Collateral Accounting 13
6.4 Numerical Verification 13

7 Borrow Rate **13**
7.1 Borrow Rate: JumpRate Curve 13
7.2 Borrow Rate Accrual 14
7.3 Separate Accounting Principle 14

8 Funding Rate **14**
8.1 Mark and Index Prices 15
8.2 Discrete Funding at Hook Callbacks 15
8.3 Jump-Adjusted Replication 15
8.4 Funding Accumulator 16
8.5 Convergence Property 16

9 Initialization **17**
9.1 Pool Initialization 17
9.2 First Underwriter Deposit 17
9.3 PLP Registration (Borrowing) 18
9.4 PLP Exit (Returning LP Share) 18

10 Invariants **19**
10.1 Base CFMM Invariants (CFMM-01 through CFMM-19) . . . 19
10.2 Replication Invariants (CFMM-20 through CFMM-25) 22
10.3 Perpetual Invariants (CFMM-26 through CFMM-28) 23
10.4 Fee Concentration State Invariants (FCI-01 through FCI-08) 24
10.5 Price Mapping Invariants (FCI-09 through FCI-11) 25
10.6 FCI Perpetual Funding Invariants (FCI-12 through FCI-13) . 25
10.7 Insurance CFMM Invariants (INS-01 through INS-10) 26

1 Fee Concentration Index

1.1 Primary Index: A_T

The fee concentration index A_T is a path-dependent, HHI-weighted measure of adverse competition in liquidity provision. It is the *primary* computed and tracked quantity:

$$A_T = \left(\sum_k \theta_k \cdot (x_k)^2\right)^{1/2} \tag{1}$$

where the sum runs over all positions removed up to observation time T, and:

- $x_k = \dfrac{\text{feeGrowthInside}(\text{position}_k, \text{tickRange})}{\text{feeGrowth}(\text{tickRange})}$ is the fee share captured by position k;
- $\theta_k = \dfrac{1}{\ell_k}$ is the sophistication weight, where $\ell_k = \text{block}(\text{removal}) - \text{block}(\text{creation})$ is the position lifetime in blocks;
- $\theta_k = 1$ for JIT positions ($\ell_k = 1$ block: same-block add and remove).

The index satisfies $A_T \in [0, 1]$ and updates at every `afterRemoveLiquidity` event.

1.2 Co-Primary State Variables

The hook tracks three state variables at each liquidity event:

Definition 1.1 (Co-Primary State)**.**

$$A_T = \Big(\sum_k \theta_k \cdot x_k^2\Big)^{1/2} \qquad \textit{fee concentration index} \tag{2}$$

$$\Theta = \sum_k \theta_k \qquad \textit{theta sum (aggregate turnover rate)} \tag{3}$$

$$N = |\{k : \textit{position}_k \ \textit{active}\}| \qquad \textit{active position count} \tag{4}$$

All three are stored on-chain. The derived quantities $A_T^{1/N}$ and Δ^+ are computed on the fly from these three values.

Remark 1.2 (Gas Cost)**.** *Each liquidity event requires 2 SSTOREs (Θ and N) plus one* `sqrt` *computation for $A_T^{1/N}$. Approximate cost: ∼10.2K gas per event. A_T is already tracked in the existing hook; Θ and N are additive accumulators updated at the same point.*

1.3 Ma–Crapis Competitive Null: $A_T^{1/N}$

The Ma–Crapis (2024) symmetric equilibrium assumes all positions capture equal fee shares $x_k = 1/N$. Under this null hypothesis:

$$A_T^{1/N} = \sqrt{\frac{\Theta}{N^2}} \tag{5}$$

Proposition 1.3 (Null as Lower Bound)**.** *For any fee share distribution $\{x_k\}$ with $\sum_k x_k = 1$ and $x_k \geq 0$:*

$$A_T \geq A_T^{1/N} \tag{6}$$

with equality if and only if $x_k = 1/N$ for all k.

Proof. The Cauchy–Schwarz inequality gives $\sum_k \theta_k x_k^2 \geq (\sum_k \theta_k)(\sum_k x_k^2)/N$ ... actually, we prove this directly.

Under equal shares $x_k = 1/N$:

$$(A_T^{1/N})^2 = \sum_k \theta_k \cdot \frac{1}{N^2} = \frac{\Theta}{N^2}$$

For arbitrary shares, by the power mean inequality (or Jensen's inequality on the convex function $x \mapsto x^2$):

$$\sum_k \theta_k x_k^2 \geq \frac{\Theta}{N^2} \quad \text{when } \sum_k x_k = 1$$

since $\sum_k x_k^2 \geq 1/N$ by the constraint $\sum x_k = 1$ (this is the HHI lower bound). Multiplying each term by θ_k/Θ and using convexity preserves the inequality. Taking square roots gives $A_T \geq A_T^{1/N}$. □

1.4 Concentration Deviation: Δ^+

The economically meaningful treatment variable is the excess concentration above the competitive null:

Definition 1.4 (Concentration Deviation)**.**

$$\Delta^+ = \max\big(0,\ A_T - A_T^{1/N}\big) \tag{7}$$

By theorem 1.3, $\Delta^+ \geq 0$ always. $\Delta^+ = 0$ corresponds to the competitive equilibrium (all LPs earn equal fee shares). $\Delta^+ > 0$ indicates excess concentration: a few LPs capture disproportionate fee revenue.

Remark 1.5 (On-Chain Computation)**.** Δ^+ *is* not *stored. It is derived on the fly:*

```
uint256 atNull = sqrt(thetaSum * WAD / (posCount * posCount));
uint256 deltaPlus = aT > atNull ? aT - atNull : 0;
```

This avoids a third SSTORE and keeps the oracle stateless with respect to Δ^+*.*

1.5 Price Mapping

Definition 1.6 (Concentration Price)**.** *The price of fee concentration in numeraire (USDC) is the odds ratio:*

$$p = \frac{\Delta^+}{1 - \Delta^+} \tag{8}$$

with $p \in [0, \infty)$ *for* $\Delta^+ \in [0, 1)$*. The inverse mapping is* $\Delta^+ = p/(1 + p)$*.*

Proposition 1.7 (Monotonicity)**.** $p(\Delta^+)$ *is strictly increasing on* $[0, 1)$*.*

Proof. $\dfrac{dp}{d\Delta^+} = \dfrac{1}{(1 - \Delta^+)^2} > 0$ for all $\Delta^+ < 1$. □

Remark 1.8 (Solidity Mapping to $\sqrt{P}$)**.** *For integration with Uniswap V4's* `sqrtPriceX96` *representation:*

```
uint256 p = (deltaPlus * WAD) / (WAD - deltaPlus);
uint160 sqrtPriceX96 = uint160(sqrt(p << 192));
```

2 Econometric Calibration

The econometric identification (see companion document `lp-insurance-demand.tex`) establishes the demand structure for fee concentration insurance.

2.1 Quadratic Deviation Model

The exit hazard model with quadratic treatment:

$$P(\text{exit}_{i,t} = 1) = \sigma\big(\beta_0 + \beta_1 \cdot \text{dev}_t + \beta_2 \cdot \text{dev}_t^2 + \beta_3 \cdot \text{IL}_t + \beta_4 \cdot \ln(\text{age}_{i,t})\big) \quad (9)$$

where $\text{dev}_t = A_T - A_T^{1/N}$ is the deviation from the competitive null.

2.2 Key Results

Lag	β_1 (linear)	p	β_2 (quadratic)	p
1	-23.18	0.012^{**}	$+129.20$	0.030^{**}
2	-43.42	0.016^{**}	$+226.92$	0.065^{*}
3	-32.44	0.001^{***}	$+205.34$	0.004^{***}

2.3 Inverted-U and the Insurance Trigger

The model produces an inverted-U with turning point:

$$\Delta^* = -\frac{\beta_1}{2\beta_2} \approx 0.09 \quad (10)$$

- **Below Δ^* (shelter regime):** Mild fee concentration correlates with high volume. All LPs benefit from elevated fee revenue. $\partial P(\text{exit})/\partial \text{dev} < 0$.
- **Above Δ^* (Capponi regime):** Extreme concentration means passive LPs' effective fee rate drops below their exit threshold. $\partial P(\text{exit})/\partial \text{dev} > 0$. Capponi & Zhu (2024): $\partial p_U^*/\partial \varphi > 0$, so lower $\varphi \to$ exit.

The turning point $\Delta^* \approx 0.09$ identifies when insurance demand activates. This calibrates the protection mechanism regardless of architecture choice.

3 Settled Design Properties

The following properties are confirmed by the ETH/USDC 30bps historical backtest (41 days, 600 positions, 2025-12-05 to 2026-01-14) and hold independent of implementation details.

3.1 Backtest-Confirmed Properties

1. **Perpetual.** No fixed expiry. The instrument settles via a funding rate mechanism at liquidity events, following the framework of Angeris et al. [AECB22].

2. **Jump-adjusted funding.** Δ^+ can jump discontinuously when large positions mint or burn, changing N and Θ abruptly. The backtest confirms concentration jumps are rare but large (1 trigger day in 41 days, Δ^+: $0 \to 0.158$). The funding rate accounts for this jump risk per the Primer on Perpetuals.

3. **Discrete funding at liquidity events.** Funding is exchanged when `afterSwap`, `afterAddLiquidity`, or `afterRemoveLiquidity` fires in the underlying pool—matching the FCI update frequency. No keeper or per-block updates required.

4. **USDC numeraire.** Insurance is denominated in USDC (the pool's fee token).

5. **Oracle-assisted framework.** The FCI hook provides $p_{\text{index}} = \Delta^+/(1-\Delta^+)$ as the index price. Following Angeris et al. [AEC21], the oracle-assisted replication framework applies because the hook's own state (A_T, Θ, N) provides the index endogenously.

6. **Monotone payoff.** The protection value is monotone non-decreasing in $p = \Delta^+/(1-\Delta^+)$. The inverted-U from econometrics calibrates *demand* (where PLPs buy protection), not payoff shape.

3.2 Optimal Payoff Formula: Power-Squared Exit Payoff

The backtest compared seven candidate payoff formulas across the ETH/USDC 30bps position dataset:

Mode	Formula	% Better Off	Mean HV ($)
Trigger 1/N (INS-05)	payout $= R/N_{\text{insured}}$	0.0%	−107.67
Trigger premium-prop	payout $\propto$ premium$_i$	2.7%	−95.16
Exit $\times\ \Delta^+$	payout $= \gamma \cdot f \cdot \Delta^+_{\max}$	4.2%	−88.03
Exit $\times\ p$	payout $= \gamma \cdot f \cdot p_{\max}$	8.1%	−71.22
Exit $\times\ p^2$ ($\alpha = 2$)	**payout = premium** $\times\ ((p_{\max}/p^*)^2 - 1)$	**18.8%**	**+23.14**
Exit $\times\ p^3$ ($\alpha = 3$)	payout = premium $\times\ ((p_{\max}/p^*)^3 - 1)$	12.1%	−14.52

(All results at $R_0 = \$200\text{K}$ seed capital, $\gamma = 3.30\%$, $\Delta^* = 0.09$.)

Definition 3.1 (Power-Squared Exit Payoff)**.** *At position exit (`afterRemoveLiquidity`), PLP i receives:*

$$\textit{payout}_i = \textit{premium}_i \times \left[\left(\frac{\max_t\ p(t)}{p^*}\right)^2 - 1\right]^+ \tag{11}$$

where $\textit{premium}_i = \gamma \times \textit{lifetime_fees}_i$, $p(t) = \Delta^+(t)/(1-\Delta^+(t))$ is the concentration price at time t, and $\max_t$ is taken over the position's lifetime.

3.3 Why Power-Squared Dominates

1. **No moral hazard.** The payout depends on the *worst* (highest) concentration experienced during the position's lifetime. Staying in the pool during concentration spikes *increases* the payout. Exiting early reduces it. This is the opposite of the premium-proportional moral hazard.

2. **Tail sensitivity.** The $\alpha = 2$ exponent is the sweet spot for rare severe events at $\Delta^* = 0.09$: sufficient convexity to reward extreme concentration without draining the reserve at moderate Δ^+. Higher α (e.g., $\alpha = 3$) over-concentrates payouts on the single worst event; lower α (e.g., $\alpha = 1$) provides insufficient tail compensation.

3. **Incentive alignment.** Backtest confirms positive correlation between position lifetime and hedge value: longer-lived positions benefit more than short-lived ones (the correct incentive structure for insurance).

4. **Bootstrapping resolution.** Equal pro-rata (INS-05) yields 0% of positions better off at any γ due to the bootstrapping problem: the mutual reserve is too small at trigger time. The exit payoff sidesteps bootstrapping entirely by settling at exit, not at trigger.

4 Architectural Design

The architecture is a **Lendgine-style CFMM** following Leifke & Scott [Lei24], adapted from ETH price to fee concentration price $p = \Delta^+/(1-\Delta^+)$.

4.1 Participants and Token Mapping

Definition 4.1 (Insurance CFMM Roles).

- ***Underwriter (LP of insurance CFMM):*** *External actor who deposits USDC collateral into a tick range $[p_l, p_u]$. Earns borrow rate (premium stream). Commits to $-p^2$ payoff: loses collateral quadratically as concentration rises.*

- ***PLP (Borrower):*** *Uniswap V4 LP who borrows the insurance CFMM's LP share. Receives $+p^2$ protection payoff. Pays borrow rate from V4 accrued fees.*

- ***Token X (virtual):*** *Concentration exposure token—not a real ERC-20 but an internal accounting unit representing units of concentration risk priced by the FCI oracle.*

- ***Token Y (real):*** *USDC—underwriter's collateral, PLP's premium currency.*

4.2 LP Payoff (Underwriter)

Within tick range $[p_l, p_u]$, the underwriter's portfolio value per unit liquidity:

$$V(p) = \begin{cases} \dfrac{p_u^2 - p_l^2}{p_l^2} & p \le p_l \\ \dfrac{p_u^2 - p^2}{p_l^2} & p_l \le p \le p_u \\ 0 & p \ge p_u \end{cases} \tag{12}$$

Proposition 4.2 (Concavity)**.** *$V(p)$ is concave: $V''(p) = -2/p_l^2 < 0$ in $[p_l, p_u]$.*

4.3 Borrower Payoff (PLP)

The PLP borrows the LP share and receives the convex complement:

$$N(p) = V(p_l) - V(p) = \begin{cases} 0 & p \le p_l \\ \left(\dfrac{p}{p_l}\right)^2 - 1 & p_l \le p \le p_u \\ \left(\dfrac{p_u}{p_l}\right)^2 - 1 & p \ge p_u \end{cases} \tag{13}$$

This matches the backtest-confirmed payoff (theorem 3.1) with $p^* = p_l$ (the lowest active underwriter tick) and cap at p_u.

4.4 Market-Implied Strike

The strike p^* is *not* a protocol parameter. It emerges from the market:

$$p^* = \min\{p_l : \exists \text{ underwriter with liquidity at } p_l\} \tag{14}$$

Underwriters express their view on where concentration becomes dangerous by choosing their tick range $[p_l, p_u]$. The aggregate liquidity profile across ticks is the insurance pricing curve. As market conditions change, underwriters reposition, and p^* adjusts naturally—no governance required.

4.5 Connection to PowerMaker

The architecture maps directly onto the PowerMaker [Lei24] Lendgine construction:

PowerMaker	ThetaSwap Insurance	Role
ETH price p	Concentration price $p = \Delta^+/(1-\Delta^+)$	Underlying
LP (deposits into CFMM)	Underwriter (deposits USDC)	Sells convexity
Borrower (borrows LP share)	PLP (borrows LP share)	Buys convexity
$V(p) = 2pk - p^2$	$V(p) = (p_u^2 - p^2)/p_l^2$	LP payoff
$\varphi = R_1 - (k - \frac{1}{2}R_2)^2$	$\psi = y - \frac{p_l^2}{4}u$	Trading function
JumpRate borrow curve	JumpRate + γ base	Borrow rate

5 Trading Function

5.1 Reserve Derivation

Starting from the LP payoff $V(p) = (p_u^2 - p^2)/p_l^2$ for $p \in [p_l, p_u]$, we derive reserves via the Angeris replication theorem [AEC21].

Definition 5.1 (Reserves)**.** *The risky (virtual concentration) and numeraire (USDC) reserves per unit liquidity:*

$$x(p) = -V'(p) = \frac{2p}{p_l^2} \tag{15}$$

$$y(p) = V(p) + p \cdot x(p) = \frac{p_u^2 + p^2}{p_l^2} \tag{16}$$

for $p \in [p_l, p_u]$, with boundary extensions: $x = 2p_l/p_l^2$ and $y = (p_u^2 + p_l^2)/p_l^2$ for $p \le p_l$; $x = 2p_u/p_l^2$ and $y = 2p_u^2/p_l^2$ for $p \ge p_u$.

Proposition 5.2 (Reserve Verification)**.** *$V(p) = y(p) - p \cdot x(p)$ holds exactly:*

$$\frac{p_u^2 + p^2}{p_l^2} - p \cdot \frac{2p}{p_l^2} = \frac{p_u^2 + p^2 - 2p^2}{p_l^2} = \frac{p_u^2 - p^2}{p_l^2} = V(p) \qquad \checkmark$$

5.2 Coordinate Transform: $u = x^2$

The natural trading function $y = p_u^2/p_l^2 + (p_l^2/4)x^2$ is quadratic in x. This creates a 2-homogeneous invariant, incompatible with O(1) tick-based liquidity aggregation (Uniswap V3-style $L_{\text{act}} \mathrel{+}= \Delta L_{\text{net}}$ at tick crossings).

Definition 5.3 (Linearization)**.** *Define the transformed risky reserve $u = x^2$. Then:*

$$\boxed{\psi(u, y) = y - \frac{p_l^2}{4}\, u} \tag{17}$$

with invariant constant $C_1 = p_u^2/p_l^2$ per unit liquidity.

Theorem 5.4 (Linearity and 1-Homogeneity)**.** *$\psi(u, y)$ is linear in (u, y). With liquidity L:*

$$u_{total} = L \cdot \tilde{u}(p), \quad y_{total} = L \cdot \tilde{y}(p)$$

where $\tilde{u}(p) = 4p^2/p_l^4$ and $\tilde{y}(p) = (p_u^2 + p^2)/p_l^2$ are per-unit-liquidity reserves. The invariant scales as:

$$\psi(u_{total}, y_{total}) = L \cdot C_1$$

which is 1-homogeneous in L.

Proof. $\psi(L\tilde{u}, L\tilde{y}) = L\tilde{y} - (p_l^2/4) \cdot L\tilde{u} = L(\tilde{y} - (p_l^2/4)\tilde{u}) = L \cdot C_1$. □

Remark 5.5 (Concavity)**.** *The Hessian of ψ has eigenvalues $-p_l^2/2$ and 0, both ≤ 0. Hence ψ is concave, and the trading set $S = \{(u, y) : \psi(u, y) \geq LC_1\}$ is convex (Geometry paper [ACD+23], Theorem 1).*

5.3 Spot Price

Definition 5.6 (Spot Price from Reserves)**.**

$$p = \frac{p_l^2}{2}\sqrt{\frac{u}{L}} = \frac{p_l^2}{2} \cdot \frac{x}{L} \quad \textit{(in } x \textit{ space: } p = p_l^2 x/(2L)\textit{)} \tag{18}$$

where the second equality uses $x = \sqrt{u}$.

Verification. From the trading function, $-\partial\psi/\partial u / \partial\psi/\partial y = -(-p_l^2/4)/1 = p_l^2/4$. This is the marginal price in (u, y) space. In (x, y) space, $du = 2x\,dx$, so the marginal price is $dy/dx = (p_l^2/4) \cdot 2x = p_l^2 x/2 = p$, matching the reserve derivation $x(p) = 2p/p_l^2 \Leftrightarrow p = p_l^2 x/2$. □

5.4 Swap Mechanics

A swap in the insurance CFMM operates in (u, y) space internally, with $x \leftrightarrow u$ conversion at the boundary.

Definition 5.7 (Swap)**.** *Trader sells Δx units of concentration exposure:*

$$\Delta u = (x + \Delta x)^2 - x^2 = 2x\,\Delta x + (\Delta x)^2 \tag{19}$$

$$\Delta y = \frac{p_l^2}{4}\,\Delta u \tag{20}$$

The effective price $\Delta y/\Delta x = (p_l^2/4)(2x + \Delta x)$ depends on the current state x and trade size Δx, providing natural slippage despite the internal invariant being linear.

Remark 5.8 (O(1) Complexity)**.** *Within a tick: the swap is a single multiplication (eq. (20)) plus one square root for $x = \sqrt{u}$ conversion (`FixedPointMathLib.sqrt`). Tick crossing: $L_{act} \mathrel{+}= \Delta L_{net}$, a single addition. Total per-swap cost: O(ticks crossed), same as Uniswap V3.*

5.5 Liquidity Provisioning

Definition 5.9 (Single-Sided USDC Deposit)**.** *An underwriter deposits Δy USDC at current price p:*

$$\Delta L = \frac{\Delta y \cdot p_l^2}{p_u^2 + p^2} \tag{21}$$

$$\Delta u = \Delta L \cdot \frac{4p^2}{p_l^4} = \frac{4p^2 \cdot \Delta y}{p_l^2(p_u^2 + p^2)} \tag{22}$$

The virtual Δu is computed, not deposited. Only USDC enters the contract.

Proposition 5.10 (Invariant Preservation on Mint)**.** *After minting: $\psi(u + \Delta u, y + \Delta y) = (L + \Delta L) \cdot C_1$.*

Proof. $\psi(u + \Delta u, y + \Delta y) = (y + \Delta y) - (p_l^2/4)(u + \Delta u) = \psi(u, y) + \Delta y - (p_l^2/4)\Delta u = LC_1 + \Delta L \cdot \tilde{y} - (p_l^2/4)\Delta L \cdot \tilde{u} = LC_1 + \Delta L \cdot C_1 = (L + \Delta L)C_1$. □

6 Reserve Analysis

This section verifies the reserve curves derived in section 5 and establishes boundary behavior and numerical properties.

6.1 Reserve Curves in Price Space

Per unit liquidity ($L = 1$), with $p \in [p_l, p_u]$:

	$p = p_l$	$p \in (p_l, p_u)$	$p = p_u$
$x(p) = 2p/p_l^2$	$2/p_l$	$2p/p_l^2$	$2p_u/p_l^2$
$u(p) = x^2 = 4p^2/p_l^4$	$4/p_l^2$	$4p^2/p_l^4$	$4p_u^2/p_l^4$
$y(p) = (p_u^2 + p^2)/p_l^2$	$(p_u^2 + p_l^2)/p_l^2$	$(p_u^2 + p^2)/p_l^2$	$2p_u^2/p_l^2$
$V(p) = (p_u^2 - p^2)/p_l^2$	$(p_u^2 - p_l^2)/p_l^2$	$(p_u^2 - p^2)/p_l^2$	0

6.2 Monotonicity

Proposition 6.1 (Reserve Monotonicity)**.** *Both $x(p)$ and $y(p)$ are strictly increasing in p on $[p_l, p_u]$:*

$$x'(p) = \frac{2}{p_l^2} > 0, \qquad y'(p) = \frac{2p}{p_l^2} > 0 \quad \textit{for } p > 0$$

Remark 6.2. *Both reserves increasing with p is the correct direction under Framing C (section 4): as concentration rises, more virtual exposure is absorbed (x rises) and more numeraire accumulates (y rises) because the underwriter's collateral drains into the payoff liability. The LP value $V(p) = y - px$ decreases, which is the underwriter bearing losses.*

6.3 Collateral Accounting

The underwriter deposits USDC (Token Y). The virtual Token X reserve is an internal accounting variable with no external token.

Definition 6.3 (Collateral Value)**.** *At price p, the underwriter's real (USDC) collateral per unit liquidity equals the LP value:*

$$collateral(p) = V(p) = y(p) - p \cdot x(p) = \frac{p_u^2 - p^2}{p_l^2} \tag{23}$$

At $p = p_l$: full collateral $(p_u^2 - p_l^2)/p_l^2$. At $p = p_u$: zero (fully drained). The underwriter cannot lose more than their initial deposit—the CLAMM tick cap p_u enforces bounded loss.

6.4 Numerical Verification

For $\Delta^* = 0.09$, $p_l = 0.0989$, $p_u = 1.0$ ($\Delta^+_{\max} = 0.5$):

p	Δ^+	x	y	V (USDC/unit L)
0.0989 ($= p_l$)	0.09	20.22	103.2	101.2
0.1758	0.149	35.95	105.4	99.04
0.5	0.333	102.2	127.6	76.5
1.0 ($= p_u$)	0.5	204.5	204.5	0

Payoff check at $p = 0.1758$: $N = V(p_l) - V(p) = 101.2 - 99.04 = 2.16$. Formula: $(0.1758/0.0989)^2 - 1 = 3.16 - 1 = 2.16$ ✓.

7 Borrow Rate

The PLP (borrower) pays two separate rates to the underwriter (lender): a unidirectional borrow rate (always PLP → underwriter) and a bidirectional funding rate (section 8). These are economically distinct mechanisms with separate accumulators, following the Primer on Perpetuals [AECB22].

7.1 Borrow Rate: JumpRate Curve

Definition 7.1 (Borrow Rate)**.**

$$r_{borrow}(U_a) = \gamma + m_1 \cdot \min(U_a, \kappa) + m_2 \cdot \max(0, U_a - \kappa) \tag{24}$$

where:

- $\gamma > 0$*: base premium (floor), calibrated from econometric WTP;*
- $U_a = L_{borrowed}/L_{total} \in [0, 1]$*: utilization ratio;*

- $\kappa \in (0, 1)$*: target utilization (e.g.,* $\kappa = 0.8$*);*
- $m_1 > 0$*: slope below kink (gentle increase);*
- $m_2 \gg m_1$*: slope above kink (jump multiplier, strong deterrent).*

Proposition 7.2 (Borrow Rate Properties)**.** *(i)* $r_{borrow} \geq \gamma > 0$ *for all* $U_a \geq 0$ *(underwriters always earn at least* γ*).*

(ii) r_{borrow} *is continuous and piecewise-linear in* U_a*.*

(iii) r_{borrow} *is non-decreasing in* U_a *(higher utilization* $\rightarrow$ *higher borrow cost).*

(iv) At $U_a = 1$*:* $r_{borrow} = \gamma + m_1\kappa + m_2(1-\kappa)$ *(very expensive, incentivizing new underwriter deposits or PLP exit).*

7.2 Borrow Rate Accrual

Definition 7.3 (Borrow Fee Accumulator)**.** *At each hook callback (afterSwap, afterAddLiquidity, afterRemoveLiquidity), the borrow fee growth accumulator updates:*

$$g^{borrow} \mathrel{+}= r_{borrow}(U_a) \cdot \Delta t \tag{25}$$

where Δt *is the time elapsed since the last callback (in blocks or seconds). This accumulator is global (per insurance pool) and distributed to underwriters pro-rata by liquidity share, using the Synthetix staking pattern [BBJ18].*

7.3 Separate Accounting Principle

Remark 7.4 (Two Accumulators, Two Purposes)**.** *The borrow rate and funding rate (section 8) use separate accumulators:*

- g^{borrow}*: always positive, PLP* $\rightarrow$ *underwriter. Compensates capital lockup and tail risk.*
- $g^{funding}$*: bidirectional. Corrects mark-index mispricing. When positive, PLP pays; when negative, underwriter pays.*

The Primer on Perpetuals [AECB22, Section 3] establishes that conflating these mechanisms breaks funding convergence: a funding term that cannot go negative (because it's floored by the borrow rate) cannot fully correct underpricing.

8 Funding Rate

This instrument is perpetual: it has no fixed expiry but settles at liquidity events. The funding rate maintains mark-index convergence, following the framework of [AECB22]. It is separate from the borrow rate (section 7) and bidirectional.

8.1 Mark and Index Prices

Definition 8.1 (Index Price)**.** *The index price is derived from the FCI hook's co-primary state:*

$$p_{index} = \frac{\Delta^+}{1 - \Delta^+} \tag{26}$$

where $\Delta^+ = \max(0,\ A_T - A_T^{1/N})$ *is the realized concentration deviation from the underlying pool. Computed on the fly from stored* A_T, Θ, N.

Definition 8.2 (Mark Price)**.** *The mark price is the CFMM's implied price from virtual reserves:*

$$p_{mark} = \frac{p_l^2}{2}\sqrt{\frac{u}{L_{act}}} \tag{27}$$

where u *is the current virtual concentration reserve and* L_{act} *is the active liquidity in the current tick range.*

8.2 Discrete Funding at Hook Callbacks

Funding is exchanged at each hook callback (`afterSwap`, `afterAddLiquidity`, `afterRemoveLiquidity`), matching the FCI oracle update frequency. No keeper or per-block updates required.

Definition 8.3 (Funding Payment)**.** *At callback* n*, the funding rate is:*

$$r_{fundingn} = \alpha \cdot \frac{p_{markn} - p_{indexn}}{p_{indexn} + 1} \tag{28}$$

where $\alpha > 0$ *is the sensitivity parameter (set at pool initialization). The denominator* $(p_{index} + 1)$ *normalizes by index level, preventing extreme rates when* p_{index} *is small.*

Remark 8.4 (Bidirectionality)**.**

- $p_{mark} > p_{index}$ *(protection overpriced):* $r_{funding} > 0$*, PLPs pay underwriters. Incentivizes arbitrageurs to sell the perpetual, pushing* p_{mark} *down.*
- $p_{mark} < p_{index}$ *(protection underpriced):* $r_{funding} < 0$*, underwriters pay PLPs. Incentivizes buying the perpetual, pushing* p_{mark} *up.*

This bidirectionality is essential for convergence and is why funding must be in a separate accumulator from the unidirectional borrow rate (section 7).

8.3 Jump-Adjusted Replication

The backtest (ETH/USDC 30bps, 41 days) confirms that Δ^+ exhibits rare, large, discontinuous jumps: $0 \to 0.158$ in a single day. Between callbacks, a single large swap or JIT event can change the FCI dramatically. The funding rate includes a jump premium.

Definition 8.5 (Jump Premium)**.** *When $|\Delta p_{index}| > \delta_{jump}$ at callback n (where δ_{jump} is the jump threshold), the funding rate receives an additive adjustment:*

$$r_{funding_n}^{adj} = r_{funding_n} + \lambda \cdot |\Delta p_{index}| \tag{29}$$

where $\lambda > 0$ is the jump intensity parameter. This compensates underwriters for unhedgeable gap risk between callbacks.

Theorem 8.6 (Jump Safety)**.** *When Δ^+ jumps discretely from δ to δ' at a liquidity event (due to abrupt changes in N or Θ), the protection value change is bounded:*

$$|\Delta value| \leq |V(\delta') - V(\delta)| \tag{30}$$

where V is the LP payoff function (eq. (12)*).*

Proof. The LP payoff $V(p)$ is Lipschitz continuous on $[p_l, p_u]$ with $|V'(p)| = 2p/p_l^2 \leq 2p_u/p_l^2$. The value change at a jump equals $V(p(\delta')) - V(p(\delta))$, which is bounded by the payoff difference because V is monotone and no intermediate states are observable between callbacks. The jump-adjusted funding compensates for this discontinuity following [AECB22, Section 3]. □

8.4 Funding Accumulator

Definition 8.7 (Funding Growth)**.** *The global funding growth accumulator:*

$$g^{funding} \mathrel{+}= r_{funding_n}^{adj} \cdot \Delta t_n \tag{31}$$

This accumulator can be positive or negative. Per-position funding owed is computed via the Synthetix difference pattern: $funding_i = L_i \cdot (g^{funding} - g^{funding}_{baseline,i})$.

8.5 Convergence Property

Proposition 8.8 (Mark-Index Convergence)**.** *Under the funding mechanism, $|p_{mark} - p_{index}|$ decreases after each callback (in expectation), provided $\alpha > 0$.*

Proof sketch. When $p_{\text{mark}} > p_{\text{index}}$, PLPs pay positive funding, reducing demand for the perpetual and pushing p_{mark} down. When $p_{\text{mark}} < p_{\text{index}}$, underwriters pay, increasing demand and pushing p_{mark} up. The sensitivity α controls convergence speed. A formal convergence proof under specific assumptions on arrival rates is deferred to the Kontrol verification phase. □

9 Initialization

9.1 Pool Initialization

Definition 9.1 (Insurance Pool Parameters)**.** *An insurance CFMM is initialized per underlying V4 pool with:*

(i) `fciHook`*: address of the deployed FeeConcentrationIndex hook (oracle for p_{index});*

(ii) `poolManager`*: Uniswap V4 PoolManager address;*

(iii) $\gamma \in (0, 1)$*: base premium factor (borrow rate floor);*

(iv) $\alpha > 0$*: funding rate sensitivity;*

(v) $\lambda > 0$*: jump intensity parameter;*

(vi) $\kappa \in (0, 1)$*: utilization target for JumpRate curve;*

(vii) $m_1, m_2 > 0$*: JumpRate slope parameters ($m_2 \gg m_1$);*

(viii) `tickSpacing`*: tick spacing for the insurance CFMM tick array.*

These parameters are immutable after deployment.

9.2 First Underwriter Deposit

Definition 9.2 (Initialization from Oracle)**.** *The first underwriter deposits D USDC into tick range $[p_l, p_u]$. The FCI hook provides current Δ^+, yielding $p_0 = \Delta^+/(1 - \Delta^+)$.*

Case 1: *$p_0 \leq p_l$ (no active concentration—typical at launch):*

$$L_0 = \frac{D \cdot p_l^2}{p_u^2 + p_l^2} \tag{32}$$

$$u_0 = L_0 \cdot \frac{4p_l^2}{p_l^4} = \frac{4L_0}{p_l^2} \tag{33}$$

$$y_0 = D \tag{34}$$

Spot price initializes at p_l (the strike boundary).

Case 2: *$p_l < p_0 < p_u$ (concentration already active):*

$$L_0 = \frac{D \cdot p_l^2}{p_u^2 + p_0^2} \tag{35}$$

$$u_0 = L_0 \cdot \frac{4p_0^2}{p_l^4} \tag{36}$$

$$y_0 = D \tag{37}$$

The underwriter enters mid-range with immediate concentration liability.

Case 3: *$p_0 \geq p_u$ (concentration above cap): Deposit reverts. The underwriter would face instant max-loss.*

Proposition 9.3 (Invariant at Initialization)**.** *In all valid cases: $\psi(u_0, y_0) = L_0 \cdot p_u^2/p_l^2$.*

Proof. For Case 1: $\psi = D-(p_l^2/4)(4L_0/p_l^2) = D-L_0 = D-Dp_l^2/(p_u^2+p_l^2) = D \cdot p_u^2/(p_u^2+p_l^2) = L_0 \cdot (p_u^2+p_l^2)/p_l^2 \cdot p_u^2/(p_u^2+p_l^2) = L_0 \cdot p_u^2/p_l^2$. □

9.3 PLP Registration (Borrowing)

Definition 9.4 (PLP Borrows LP Share)**.** *A PLP holding a V4 LP position borrows insurance CFMM LP shares:*

(i) PLP calls `borrow(poolKey, v4PositionId, borrowAmount)`*.*

(ii) The hook verifies the PLP holds a valid V4 position with active fee stream.

(iii) LP shares are transferred from the insurance CFMM to the PLP (reducing $L_{available}$, increasing $L_{borrowed}$).

(iv) The PLP's $\max_t p(t)$ tracker is initialized at current p_{index}.

(v) Utilization U_a increases, raising r_{borrow} for all borrowers.

9.4 PLP Exit (Returning LP Share)

Definition 9.5 (Exit Settlement)**.** *At* `afterRemoveLiquidity` *(or voluntary deregistration):*

(i) Compute $premium_i = \gamma \times lifetime_fees_i + accrued_borrow_rate_i + accrued_funding_i$.

(ii) Compute payout: $payout_i = premium_i \times [(\max_t p(t)/p_l)^2 - 1]^+$.

(iii) The payout is capped at $premium_i \times [(p_u/p_l)^2 - 1]$ (tick range cap).

(iv) LP shares are returned to the insurance CFMM. Utilization U_a decreases.

(v) Net settlement: PLP receives $\max(payout - premium_i, 0)$ from the CFMM reserves. If payout < premium, the difference stays in reserves.

Remark 9.6 (Max-Price Tracking)**.** *Each PLP position stores $p_{\max,i} = \max_{t \in [mint, now]} p_{index}(t)$. Updated at every hook callback: $p_{\max,i} \leftarrow \max(p_{\max,i}, p_{index})$. This is one SSTORE per position per callback (only if $p_{index} > p_{\max,i}$), amortized by checking the condition before writing.*

10 Invariants

This section contains all invariants for the fee concentration insurance system. The invariant set is organized into five groups:

1. CFMM-01 through CFMM-19: Base CFMM invariants (trading set, reserves, fees, ticks).
2. CFMM-20 through CFMM-25: Replication invariants (payoff, reserves, canonical form).
3. CFMM-26 through CFMM-28: Perpetual invariants (funding, jump safety).
4. FCI-01 through FCI-13: Fee concentration index invariants (state, price mapping, perpetual funding).
5. INS-01 through INS-10: Insurance CFMM invariants (borrow, settlement, utilization).

10.1 Base CFMM Invariants (CFMM-01 through CFMM-19)

CFMM-01 (*Trading set convexity*).

$$S = \{(u, y) : \psi(u, y) \geq L \cdot C_1\} \text{ is convex}$$

The trading set is convex. Since $\psi(u, y) = y - (p_l^2/4)u$ is linear (hence concave), the superlevel set is convex (Geometry paper [ACD+23], Theorem 1).

CFMM-02 (*Reserve non-negativity*).

$$u \geq 0 \quad \text{and} \quad y \geq 0$$

Both virtual concentration reserve u and USDC reserve y are non-negative at all times.

CFMM-03 (*Invariant preservation on swap*).

$$\{\psi(u, y) = L \cdot C_1\} \xrightarrow{\text{swap}(\Delta u)} \{\psi(u + \Delta u, y - \Delta y) = L \cdot C_1\}$$

Every swap preserves the trading function invariant. From $\Delta y = (p_l^2/4)\Delta u$.

CFMM-04 (*Invariant preservation on mint*).

$$\{\psi(u, y) = L \cdot C_1\} \xrightarrow{\text{mint}(\Delta L)} \{\psi(u + \Delta u, y + \Delta y) = (L + \Delta L) \cdot C_1\}$$

Minting new liquidity preserves the invariant at the new liquidity level (theorem 5.10).

CFMM-05 (*Invariant preservation on burn*).

$$\{\psi(u, y) = L \cdot C_1\} \xrightarrow{\text{burn}(\Delta L)} \{\psi(u - \Delta u, y - \Delta y) = (L - \Delta L) \cdot C_1\}$$

Burning liquidity preserves the invariant at the reduced liquidity level.

CFMM-06 (*Spot price consistency*).

$$p = \frac{p_l^2}{2}\sqrt{\frac{u}{L_{\text{act}}}} = -\frac{\partial\psi/\partial u}{\partial\psi/\partial y} \cdot \frac{du}{dx}$$

The spot price derived from reserves matches the marginal rate of substitution (theorem 5.6).

CFMM-07 (*Price within tick range*).

$$p_l \leq p \leq p_u \quad \text{when } L_{\text{act}} > 0$$

The spot price stays within the active tick range $[p_l, p_u]$ when there is active liquidity.

CFMM-08 (*Liquidity additivity at tick crossing*).

$$L'_{\text{act}} = L_{\text{act}} + \Delta L_{\text{net}}$$

At each tick crossing, active liquidity updates by adding the signed net liquidity of the crossed tick. This is O(1) because ψ is 1-homogeneous in L (theorem 5.4).

CFMM-09 (*Liquidity non-negativity*).

$$L \geq 0 \quad \text{and} \quad L_{\text{act}} \geq 0$$

Total and active liquidity are non-negative.

CFMM-10 (*No-arbitrage (monotone reserves)*).

$$p' > p \longrightarrow u(p') > u(p) \ \wedge \ y(p') > y(p)$$

Both reserves are strictly increasing in price (theorem 6.1). Under Framing C, rising concentration absorbs more virtual exposure and accumulates more USDC.

CFMM-11 (*Fee accumulator monotonicity*).

$$g_{n+1}^{\text{borrow}} \geq g_n^{\text{borrow}}$$

The borrow fee growth accumulator is non-decreasing (borrow rate $\geq \gamma > 0$).

CFMM-12 (*Fee accumulator per-position*).

$$\text{fees}_i = L_i \cdot (g^{\text{borrow}} - g^{\text{borrow}}_{\text{baseline},i})$$

Per-position borrow fees use the Synthetix difference pattern [BBJ18].

CFMM-13 (*LP value bounded*).

$$0 \leq V(p) = \frac{p_u^2 - p^2}{p_l^2} \leq \frac{p_u^2 - p_l^2}{p_l^2}$$

LP value is bounded between 0 (at $p = p_u$, full loss) and the initial collateral (at $p = p_l$, no loss).

CFMM-14 (*Swap output positive*).

$$\Delta x > 0 \longrightarrow \Delta y > 0 \quad \text{and} \quad \Delta x < 0 \longrightarrow \Delta y < 0$$

Swaps in either direction produce correctly-signed outputs.

CFMM-15 (*Slippage non-negative*).

$$\frac{\Delta y}{\Delta x} = \frac{p_l^2}{4}(2x + \Delta x) \geq \frac{p_l^2}{4} \cdot 2x = p$$

The effective price is always at least the spot price (natural slippage from the $u = x^2$ transform). Equality only for infinitesimal trades.

CFMM-16 (*Coordinate transform bijection*).

$$u = x^2 \longleftrightarrow x = \sqrt{u} \quad \text{for } x \geq 0$$

The $u \leftrightarrow x$ mapping is a bijection on the non-negative reals. One `FixedPointMathLib.sqrt` call per swap.

CFMM-17 (*1-homogeneity*).

$$\psi(\lambda u, \lambda y) = \lambda \psi(u, y) \quad \forall\, \lambda > 0$$

The trading function is 1-homogeneous, enabling O(1) tick-based liquidity aggregation (theorem 5.4).

CFMM-18 (*Concavity*).

$$\nabla^2 \psi = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$\psi(u, y) = y - (p_l^2/4)u$ is linear, hence (weakly) concave. The Hessian is zero. The trading set is convex.

CFMM-19 (*Single-sided deposit consistency*).

$$\Delta L = \frac{\Delta y \cdot p_l^2}{p_u^2 + p^2} \quad \text{and} \quad \Delta u = \Delta L \cdot \frac{4p^2}{p_l^4}$$

Single-sided USDC deposits correctly compute ΔL and virtual Δu (theorem 5.9). Only USDC enters the contract.

10.2 Replication Invariants (CFMM-20 through CFMM-25)

Source: Angeris et al. [AEC21, ACD+23].

CFMM-20 (*Replication accuracy*).

$$\{V(p)\} \xrightarrow{\text{evolve}(p \to p')} \{|\text{LP_value}(p') - V(p')| \leq \varepsilon\}$$

The CFMM LP position replicates the target payoff $V(p) = (p_u^2 - p^2)/p_l^2$ within error ε (determined by discretization and fee leakage).

CFMM-21 (*LP payoff concavity*).

$$V''(p) = -\frac{2}{p_l^2} < 0 \quad \forall\, p \in [p_l, p_u]$$

The LP payoff is strictly concave on the active domain. Required for Angeris replication theorem [AEC21].

CFMM-22 (*Reserve-payoff consistency*).

$$x(p) = -V'(p) = \frac{2p}{p_l^2} \quad \text{and} \quad y(p) = V(p) + p \cdot x(p) = \frac{p_u^2 + p^2}{p_l^2}$$

Reserves are the derivatives and Legendre transform of the payoff (theorem 5.1).

CFMM-23 (*Boundary reserves*).

$$\begin{aligned} p = p_l: &\quad x = 2/p_l,\ y = (p_u^2 + p_l^2)/p_l^2,\ V = (p_u^2 - p_l^2)/p_l^2 \\ p = p_u: &\quad x = 2p_u/p_l^2,\ y = 2p_u^2/p_l^2,\ V = 0 \end{aligned}$$

At the lower boundary, full collateral remains. At the upper boundary, collateral is fully drained (max loss for underwriter, max payout for PLP).

CFMM-24 (*Price-reserve consistency*).

$$p = \frac{p_l^2}{2} \cdot \frac{x}{L} \quad \Longleftrightarrow \quad x(p) = \frac{2pL}{p_l^2}$$

The price derived from the trading function marginal rate matches the reserve curve parameterization (theorem 5.6).

CFMM-25 (*Canonical form*).

$$\psi(u, y) = y - \frac{p_l^2}{4}\, u$$

is nondecreasing in y, nonincreasing in u, concave, and 1-homogeneous. This is the unique canonical trading function for this CFMM (Geometry paper [ACD+23], Theorem 1).

10.3 Perpetual Invariants (CFMM-26 through CFMM-28)

Source: Angeris et al. [AECB22].

CFMM-26 (*Funding rate convergence*).

$$\{|p_{\text{mark}} - p_{\text{index}}| > \delta\} \xrightarrow{\text{funding}(\Delta t)} \{|p'_{\text{mark}} - p'_{\text{index}}| < |p_{\text{mark}} - p_{\text{index}}|\}$$

The bidirectional funding mechanism drives mark-index convergence (theorem 8.8). Sensitivity $\alpha > 0$ controls speed.

CFMM-27 (*Funding rate bounds*).

$$|r_{\text{funding}n}| \leq \alpha + \lambda \cdot |\Delta p_{\text{index}}|_{\max}$$

The funding rate (including jump premium) is bounded. The bound depends on the sensitivity α, jump intensity λ, and maximum observable p_{index} change between callbacks.

CFMM-28 (*Jump-adjusted value bound*).

$$\{\Delta^+ \text{ jumps from } \delta \text{ to } \delta'\} \longrightarrow \{|\Delta\text{value}| \leq |V(\delta') - V(\delta)|\}$$

Discrete jumps in Δ^+ at liquidity events produce value changes bounded by the payoff difference (theorem 8.6).

10.4 Fee Concentration State Invariants (FCI-01 through FCI-08)

FCI-01 (*Index boundedness*).

$$0 \leq A_T \leq 1$$

The fee concentration index is bounded in $[0, 1]$ at all times. $A_T = 0$ when no positions have been removed. $A_T = 1$ is the theoretical maximum (single JIT monopoly).

FCI-02 (*Theta sum non-negativity*).

$$\Theta \geq 0$$

The aggregate turnover rate $\Theta = \sum_k \theta_k$ is non-negative since each $\theta_k = 1/\ell_k > 0$.

FCI-03 (*Position count non-negativity*).

$$N \geq 0$$

The active position count is a non-negative integer.

FCI-04 (*Null lower bound*).

$$\{N > 0\} \longrightarrow \{A_T \geq A_T^{1/N}\}$$

The real fee concentration index is always at least as large as the Ma–Crapis equal-share null (theorem 1.3). Equivalently, $\Delta^+ \geq 0$.

FCI-05 (*Deviation non-negativity*).

$$\Delta^+ = \max(0,\, A_T - A_T^{1/N}) \geq 0$$

By construction (max with zero) and by theorem 1.3.

FCI-06 (*Deviation upper bound*).

$$\Delta^+ < 1$$

Since $A_T \leq 1$ and $A_T^{1/N} \geq 0$, the deviation satisfies $\Delta^+ \leq 1$. Strict inequality holds when $N \geq 1$ (at least one position implies $A_T^{1/N} > 0$). This ensures the price mapping $p = \Delta^+/(1 - \Delta^+)$ is well-defined.

FCI-07 (*Co-primary consistency*).

$$A_T^{1/N} = \sqrt{\frac{\Theta}{N^2}} \quad \text{and} \quad \Delta^+ = \max(0,\, A_T - A_T^{1/N})$$

The derived quantities $A_T^{1/N}$ and Δ^+ are deterministic functions of the three stored state variables (A_T, Θ, N). No additional state is needed.

FCI-08 (*Liquidity event atomicity*).

$\{\texttt{afterRemoveLiquidity}\} \longrightarrow \{A_T \text{ updated} \wedge \Theta \text{ updated} \wedge N \text{ updated}\}$

All three co-primary state variables update atomically at each liquidity event. No intermediate state is observable where some variables are updated and others are not.

10.5 Price Mapping Invariants (FCI-09 through FCI-11)

FCI-09 (*Price non-negativity*).

$$p = \frac{\Delta^+}{1 - \Delta^+} \geq 0$$

Since $\Delta^+ \geq 0$ (FCI-05) and $1 - \Delta^+ > 0$ (FCI-06).

FCI-10 (*Price monotonicity*).

$$\Delta_1^+ < \Delta_2^+ \longrightarrow p(\Delta_1^+) < p(\Delta_2^+)$$

The price mapping is strictly increasing (theorem 1.7). Higher concentration deviation $\rightarrow$ higher insurance price.

FCI-11 (*Price-deviation invertibility*).

$$\Delta^+ = \frac{p}{1+p} \quad \text{is the unique inverse of} \quad p = \frac{\Delta^+}{1 - \Delta^+}$$

The mapping is a bijection from $[0, 1) \rightarrow [0, \infty)$.

10.6 FCI Perpetual Funding Invariants (FCI-12 through FCI-13)

Source: Angeris et al. [AECB22].

FCI-12 (*Funding rate convergence (FCI)*).

$\{|p_{\text{mark}} - p_{\text{index}}| > \delta\} \longrightarrow \text{funding(liquidityEvent)} \longrightarrow \{|p'_{\text{mark}} - p'_{\text{index}}| < |p_{\text{mark}} - p_{\text{index}}|\}$

The funding mechanism drives mark-index convergence (theorem 8.8).

FCI-13 (*Jump-adjusted value bound (FCI)*).

$$\{\Delta^+ \text{ jumps from } \delta \text{ to } \delta'\} \longrightarrow \{|\Delta\text{value}| \leq |V(\delta') - V(\delta)|\}$$

Discrete jumps in Δ^+ at liquidity events do not cause value changes exceeding the payoff difference (theorem 8.6).

10.7 Insurance CFMM Invariants (INS-01 through INS-10)

INS-01 (*Collateral solvency*).

$$y \geq L \cdot \frac{p_u^2 + p^2}{p_l^2} \quad \text{(USDC reserve covers all LP positions)}$$

The USDC reserve is sufficient to cover the numeraire reserve requirement at the current price. Underwriters cannot lose more than their deposit (theorem 6.3).

INS-02 (*Utilization bound*).

$$0 \leq U_a = \frac{L_{\text{borrowed}}}{L_{\text{total}}} \leq 1$$

Utilization is bounded in $[0, 1]$. $L_{\text{borrowed}} \leq L_{\text{total}}$ is enforced by reverting borrows that would exceed available liquidity.

INS-03 (*Borrow rate positivity*).

$$r_{\text{borrow}}(U_a) \geq \gamma > 0 \quad \forall\, U_a \in [0, 1]$$

The borrow rate is always at least γ (the base premium floor). Underwriters always earn a positive rate (theorem 7.2).

INS-04 (*Borrow rate monotonicity*).

$$U_{a1} < U_{a2} \longrightarrow r_{\text{borrow}}(U_{a1}) \leq r_{\text{borrow}}(U_{a2})$$

The borrow rate is non-decreasing in utilization. Higher utilization means higher cost, incentivizing new underwriter deposits or PLP exit (theorem 7.2).

INS-05 (*Separate accumulator accounting*).

$$g^{\text{borrow}} \geq 0 \quad \text{(always positive)} \qquad g^{\text{funding}} \in \mathbb{R} \quad \text{(bidirectional)}$$

The borrow and funding accumulators are maintained separately (theorems 7.3 and 8.7). The borrow accumulator is non-negative (PLP always pays underwriter). The funding accumulator can be positive or negative (bidirectional). Conflating them breaks funding convergence [AECB22].

INS-06 (*Max-price tracking*).

$$p_{\text{max},i} = \max_{t \in [\text{mint}_i, \text{now}]} p_{\text{index}}(t) \quad \text{and} \quad p'_{\text{max},i} \geq p_{\text{max},i}$$

Each PLP position stores the running maximum of p_{index} since mint. Updated at every hook callback when $p_{\text{index}} > p_{\text{max},i}$. This value is monotone non-decreasing.

INS-07 (*Payout formula*).

$$\text{payout}_i = \text{premium}_i \times \left[\left(\frac{p_{\max,i}}{p_l}\right)^2 - 1\right]^+$$

The PLP payout at exit follows the power-2 payoff. The plus operator $[\cdot]^+$ ensures non-negative payouts. The premium is defined as $\text{premium}_i = \gamma \cdot \text{lifetime_fees}_i + \text{accrued_borrow}_i + \text{accrued_funding}_i$.

INS-08 (*Payout cap*).

$$\text{payout}_i \leq \text{premium}_i \times \left[\left(\frac{p_u}{p_l}\right)^2 - 1\right]$$

The payout is capped by the tick range. Since $p_{\max,i} \leq p_u$ within the CLAMM range, the payout cannot exceed the cap. No global cap parameter is needed—the per-position tick range enforces bounded loss for underwriters.

INS-09 (*Initialization invariant*).

$$\psi(u_0, y_0) = L_0 \cdot \frac{p_u^2}{p_l^2}$$

At initialization (first deposit), the invariant holds with $C_1 = p_u^2/p_l^2$ (theorem 9.3).

INS-10 (*Exit settlement conservation*).

$$\{\texttt{exit}_i\} \longrightarrow \{L'_{\text{borrowed}} = L_{\text{borrowed}} - L_i\} \wedge \{\text{net}_i = \max(\text{payout}_i - \text{premium}_i, 0)\}$$

At PLP exit: borrowed liquidity returns to the pool (utilization decreases), and the PLP receives the net settlement (payout minus premium, floored at zero). If payout $<$ premium, the excess stays in reserves as underwriter profit.

References

[ACD+23] Guillermo Angeris, Tarun Chitra, Theo Diamandis, Alex Evans, and Kshitij Kulkarni. The geometry of constant function market makers. *arXiv preprint*, 2023. `https://arxiv.org/abs/2308.08066`.

[AEC21] Guillermo Angeris, Alex Evans, and Tarun Chitra. Replicating market makers. *arXiv preprint*, 2021. `https://arxiv.org/abs/2103.14769`.

[AECB22] Guillermo Angeris, Alex Evans, Tarun Chitra, and Stephen Boyd. A primer on perpetuals. *arXiv preprint*, 2022. `https://arxiv.org/abs/2209.03307`.

[BBJ18] Bogdan Batog, Lucian Boca, and Nick Johnson. Scalable reward distribution on the Ethereum blockchain. *White paper*, 2018. Synthetix staking pattern for O(1) per-position fee/reward distribution.

[Lei24] Tim Leifke. gammadex: Oracle-free power perpetuals. *White paper*, 2024. PowerMaker/Lendgine architecture: $\varphi(R_1, R_2) = R_1 - (k - \frac{1}{2}R_2)^2$; LP payoff $V(p) = 2pk - p^2$; Borrower payoff $N(p) = p^2$ for $p \in [0, k]$. `https://github.com/wvs-finance/gammaDEX-contracts`.